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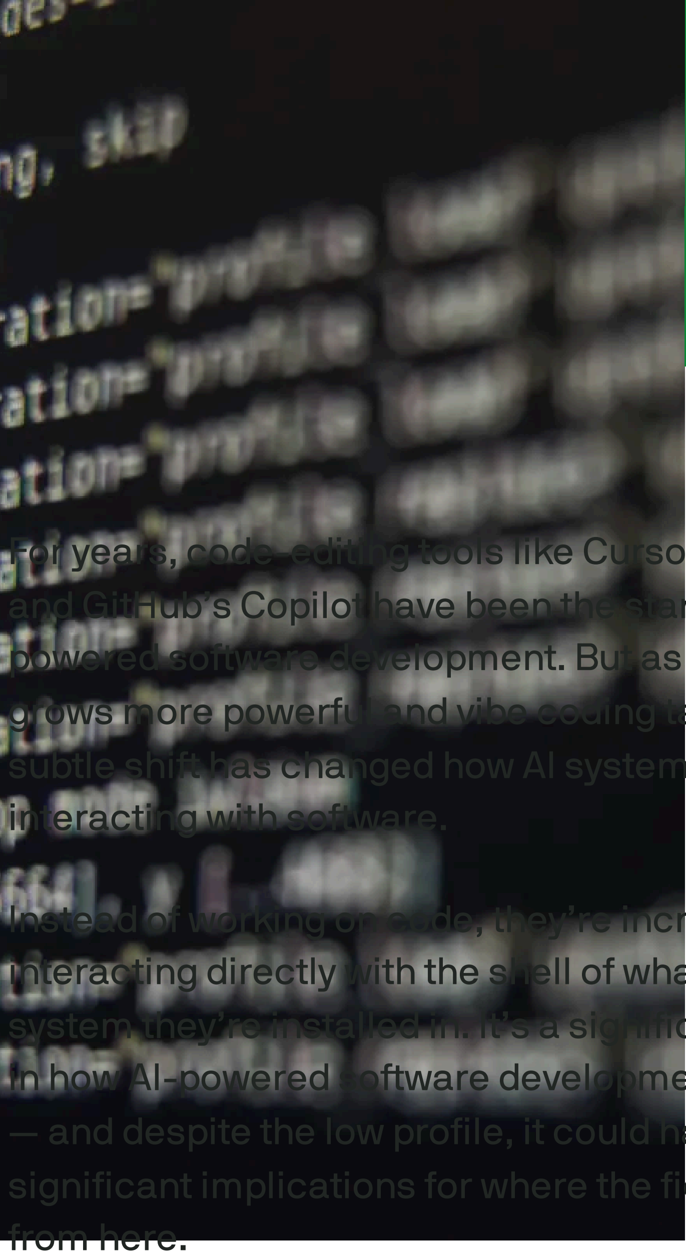
AI

AI coding tools are shifting to a surprising place: The terminal

Russell Brandom — 9:30 AM PDT · July 15, 2025



For years, code-editing tools like Cursor, Windsurf, and GitHub's Copilot have been the standard for AI-powered software development. But as agentic AI grows more powerful and vibe coding takes off, a subtle shift has changed how AI systems are interacting with software.

Instead of working on code, they're increasingly interacting directly with the shell of whatever system they're installed in. It's a significant change in how AI-powered software development happens — and despite the low profile, it could have significant implications for where the field goes from here.

IMAGE CREDITS: INIMMA-IS / GETTY IMAGES

The terminal is best known as the black-and-white screen you remember from '90s hacker movies — a very old-school way of running programs and manipulating data. It's not as visually impressive as contemporary code editors, but it's an extremely powerful interface if you know how to use it. And while code-based agents can write and debug code, terminal tools are often needed to get software from written code to something that can actually be used.

The clearest sign of the shift to the terminal has come from major labs. Since February, Anthropic, DeepMind, and OpenAI have all released command-line coding tools (Claude Code, Gemini CLI, and CLI Codex, respectively), and they're

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That shift has been easy to miss, since they're largely operating under the same branding as previous coding tools. But under the hood, there have been real changes in how agents interact with other computers, both online and offline. Some believe those changes are just getting started.

"Our big bet is that there's a future in which 95% of LLM-computer interaction is through a terminal-like interface," says Mike Merrill, co-creator of the leading terminal-focused benchmark Terminal-Bench.

Terminal-based tools are also coming into their own just as prominent code-based tools are starting to look shaky. The AI code editor Windsurf has been torn apart by dueling acquisitions, with senior executives hired away by Google and the remaining company acquired by Cognition — leaving the consumer product's long-term future uncertain.

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At the same time, new research suggests programmers may be overestimating productivity gains from conventional tools. A METR study testing Cursor Pro, Windsurf's main competitor, found that while developers estimated they could complete tasks 20% to 30% faster, the observed process was nearly 20% slower. In short, the code assistant was actually costing programmers time.

That has left an opening for companies like Warp, which currently holds the top spot on Terminal-Bench. Warp bills itself as an "agentic development environment," a middle ground between IDE programs and command-line tools like Claude Code.

But Warp founder Zach Lloyd is still bullish on the terminal, seeing it as a way to tackle problems that would be out of scope for a code editor like Cursor.

"The terminal occupies a very low level in the developer stack, so it's the most versatile place to be running agents," Lloyd says.

To understand how the new approach is different, it can be helpful to look at the benchmarks used to measure them. The code-based generation of tools was focused on solving GitHub issues, the basis of the SWE-Bench test. Each problem on SWE-Bench is an open issue from GitHub — essentially, a piece of code that doesn't work.

Models iterate on the code until they find something that works, solving the problem. Integrated products like Cursor have built more sophisticated approaches to the problem, but the GitHub/SWE-Bench model is still the core of how these tools approach the problem: starting with broken code and turning it into code that works.

Terminal-based tools take a wider view, looking beyond the code to the whole environment a program is running in. That includes coding but also more DevOps-oriented tasks like configuring a Git server or troubleshooting why a script won't run.

In one TerminalBench problem, the instructions give a decompression program and a target text file, challenging the agent to reverse-engineer a matching compression algorithm. Another asks the agent to build the Linux kernel from source, failing to mention that the agent will have to download the source code itself. Solving the issues requires the kind of bull-headed problem-solving ability that programmers need.

"What makes TerminalBench hard is not just the questions that we're giving the agents," says Terminal-Bench co-creator Alex Shaw. "It's the environments that we're placing them in."

Crucially, this new approach means tackling a problem step-by-step — the same skill that makes agentic AI so powerful. But even state-of-the-art agentic models can't handle all of those environments. Warp earned its high score on Terminal-Bench by solving just over half of the problems — a mark of how challenging the benchmark is and how much work still needs to be done to unlock the terminal's full potential.

Still, Lloyd believes we're already at a point where terminal-based tools can reliably handle much of a developer's non-coding work — a value proposition that's hard to ignore.

"If you think of the daily work of setting up a new project, figuring out the dependencies and getting it runnable, Warp can pretty much do that autonomously," says Lloyd. "And if it can't do it, it will tell you why."

Topics: AI AI coding tools code editor

IMAGE CREDITS: JAKUB PORZYCKI/NURPHOTO / GETTY IMAGES

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OpenAI debuts Codex CLI, an open source coding tool for terminals

Kyle Wiggers — 10:00 AM PDT · April 16, 2025

In a bid to inject AI into more of the programming process, OpenAI is launching Codex CLI, a coding "agent" designed to run locally from terminal software.

Announced on Wednesday alongside OpenAI's newest AI models, o3 and o4-mini, Codex CLI links OpenAI's models with local code and computing tasks, OpenAI says. Via Codex CLI, OpenAI's models can write and edit code on a desktop and take certain actions, like moving files.

Codex CLI appears to be a small step in the direction of OpenAI's broader agentic coding vision. Recently, the company's CFO, Sarah Friar, described what she called the "agentic software engineer," a set of tools OpenAI intends to build that can take a project description for an app and effectively create it and even quality assurance test it.

Codex CLI won't go that far. But it *will* integrate OpenAI's models, eventually including o3 and o4-mini, with the clients that process code and computer commands, otherwise known as command-line interfaces.

It's also open source, OpenAI says.

"[Codex CLI is] a lightweight, open source coding agent that runs locally in your terminal," an OpenAI spokesperson told TechCrunch via email. "The goal [is to] give users a minimal, transparent interface to link models directly with [code and tasks]."

In a blog post provided to TechCrunch, OpenAI added, "You can get the benefits of multimodal reasoning from the command line by passing screenshots or low fidelity sketches to the model, combined with access to your code locally [via Codex CLI]."

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To spur use of Codex CLI, OpenAI plans to dole out $1 million in API grants to eligible software development projects. The company says it'll award $25,000 blocks of API credits to the projects chosen.

AI coding tools come with risks, of course. Many studies have shown that code-generating models frequently fail to fix security vulnerabilities and bugs, and even introduce them. It's best to keep that in mind before giving AI access to sensitive files or projects — must less entire systems.

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Kyle Wiggers

AI Editor

Kyle Wiggers was TechCrunch's AI Editor until June 2025. His writing has appeared in VentureBeat and Digital...

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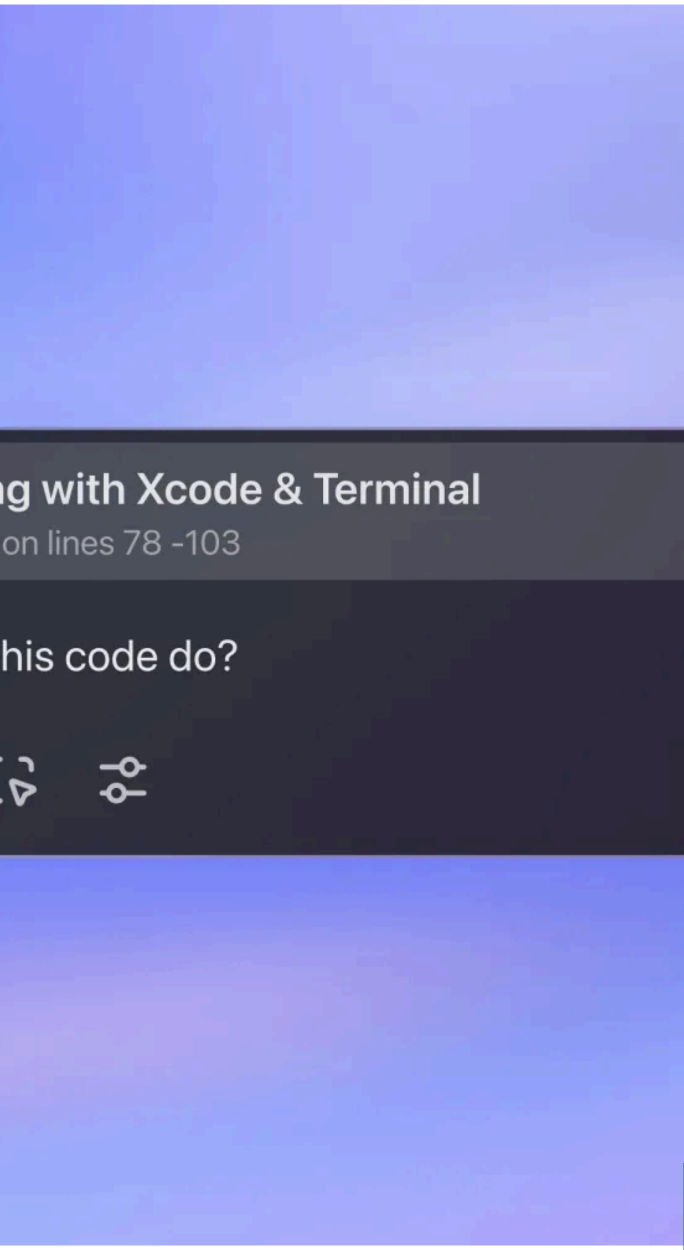

IMAGE CREDITS: OPENAI

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ChatGPT can now read some of your Mac's desktop apps

Maxwell Zeff — 10:00 AM PST · November 14, 2024

OpenAI's ChatGPT is starting to work with other apps on your computer.

On Thursday, the startup announced the ChatGPT desktop app for macOS can now read code in a

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handful of developer-focused coding apps, such as VS Code, Xcode, TextEdit, Terminal, and iTerm2.

That means developers will no longer have to copy and paste their code into ChatGPT, which has become a common way to use the chatbot. Now when the feature is enabled, OpenAI will automatically send the section of code you're working on through its chatbot as context, alongside your prompt.

However, unlike popular AI coding tools such as Cursor or GitHub Copilot, ChatGPT is currently unable to write code directly into developer apps on your behalf.

The feature, called Work with Apps, is far from an AI agent, but OpenAI says getting ChatGPT to understand other apps is a "key building block" toward building agentic systems. One of the biggest challenges facing AI agents today is getting them to understand the rest of your computer screen, as opposed to prompts or their own responses.

OpenAI says it's focusing this feature on coding apps to start; this is likely because AI coding assistants have taken off as one of the most popular use cases for LLMs. The feature is available to Plus and Teams users today and will roll out to Enterprise and Edu in the next few weeks. OpenAI says ChatGPT will be able to work with other types of apps moving forward, specifically text-based apps that could be used for writing tasks.

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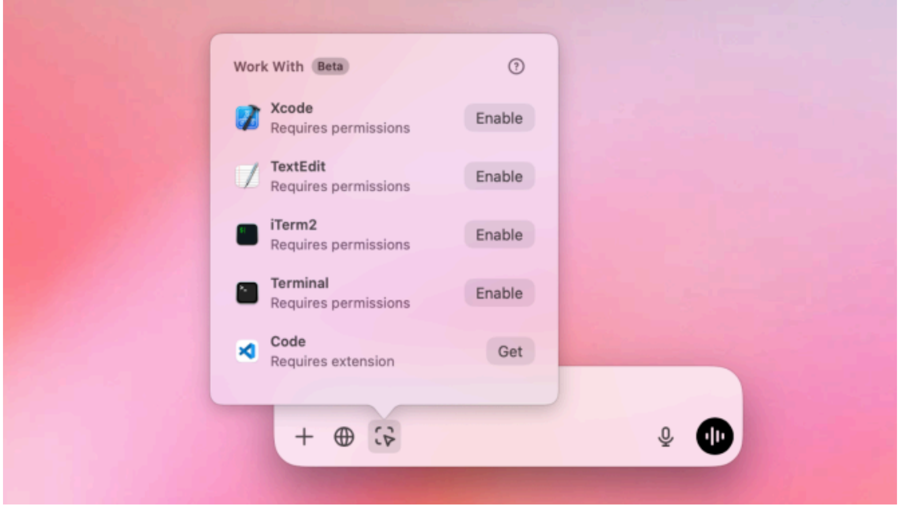

YOU CAN NOW SELECT A FEW CODING APPS FOR CHATGPT TO WORK WITH.

IMAGE CREDITS:OPENAI

In a demo with TechCrunch, an OpenAI employee opened the ChatGPT app and an Xcode environment containing a simple project modeling the solar system — although it was missing Earth. The employee selected an Xcode tab within ChatGPT, which tells the AI chatbot to look at the app, and prompted the chatbot to "add the missing planets." The chatbot was able to complete the task, writing a line of code to represent Earth that matched the rest of the project's format. They still had to paste ChatGPT's answer back into their environment, though.

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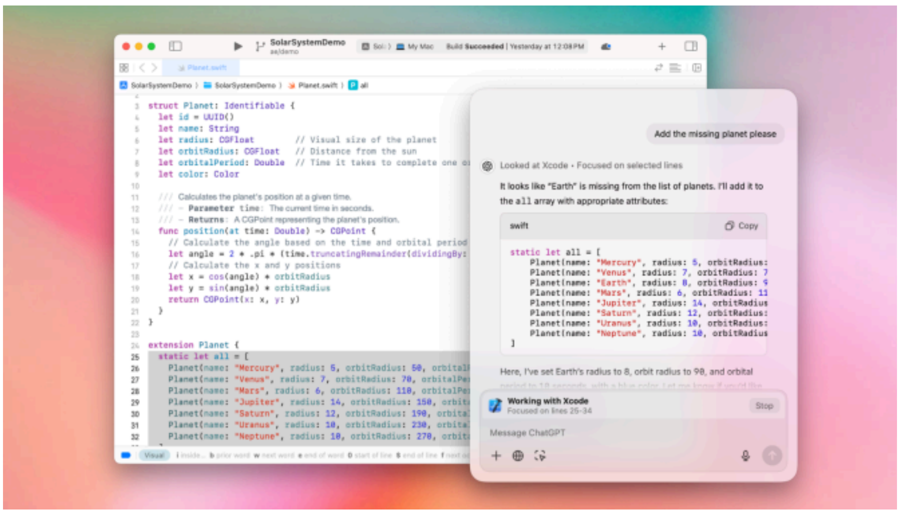
In order to read different apps, OpenAI is mostly relying on the macOS accessibility API to read text and translate it to ChatGPT, according to OpenAI desktop product lead Alexander Embiricos. The screen reader on macOS, which helps [Apple's VoiceOver feature] work, has been around for nearly two decades. It's generally considered pretty reliable for most common apps, but not everything.

For some apps, such as Microsoft's VS Code, Work with Apps requires users to install a special extension to query content. And, as the name suggests, Apple's screen reader can only read text, so it can't help ChatGPT understand visual elements, such as photos, the orientation of objects, or videos.

Work with Apps will send your last 200 lines of code through ChatGPT alongside every prompt for certain apps. For others, all the code in your foremost window will be used as input for the chatbot. You can highlight sections of code or text to help ChatGPT focus on the right part of the project, but ChatGPT will also include text surrounding it. This all sounds like it will use a lot of input tokens.



CHATGPT WORKING WITH XCODE. IMAGE CREDITS:OPENAI

It’s unclear how OpenAI plans to branch this feature out to other apps that are not compatible with Apple’s screen reader. Anthropic, one of OpenAI’s competitors, released an AI system that analyzes screenshots of a user’s desktop to understand and use other apps. To be frank, Anthropic’s approach leaves a lot to be desired in its current state. It’s slow and makes a lot of mistakes. However, it’s a more general-purpose version of an AI agent that doesn’t rely on APIs and can do more than just read text in another window.

“This isn’t meant to be an agent; it’s a way to collaborate with coding tools to start, and there will be more tools coming soon” said Embiricos in a briefing with TechCrunch. “On the side of agents, I think this is a really key building block. This idea that ChatGPT understands or can work with all the content that you have so that it can help with it.”

This step toward agents is especially notable given recent reports that OpenAI is nearing the release of a general-purpose AI agent, codenamed “Operator,” according to Bloomberg. The tool is expected to arrive in early 2025 and would rival other early attempts at general-purpose AI agents, such as Anthropic’s Computer use or Google’s reported “Jarvis” agent.

OpenAI is first releasing these features on macOS, shortly before Apple launches an integration with ChatGPT in December. It’s unclear when Work with Apps will come to Windows, the operating system created by OpenAI’s largest backer, Microsoft.

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